

PRADEEP RAJASEKAR

AI Research Engineer

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PROFESSIONAL SUMMARY

AI Research Engineer with 4+ years of experience across computer vision, LLMs, agentic AI, and multi-modal systems rooted in deep mathematical foundations (linear algebra, probabilistic modeling, geometric vision) that enable rapid domain transfer. Built production-scale biomechanical AI for markerless motion analysis (60% error reduction), architected agentic LLM coaching workflows with domain-specific fine-tuning, and shipped real-time inference on Apple CoreML, NVIDIA TensorRT, and GCP. Research-to-production leader: NVIDIA GTC 2025 presenter, TAITRA “Go Healthy Taiwan” award-winning proposal author, and Taiwan Expo Europe engineering representative. Drives cross-functional R&D, establishes organizational AI coding standards, and mentors engineering teams from concept through deployment.

CORE COMPETENCIES

Mathematical Foundations: Linear Algebra, Calculus, Probabilistic Modeling, Multi-View Geometry, Temporal Modeling, Geometric Vision, Optimization Theory

Deep Learning & Models: Transformers, Diffusion Models, GANs, Attention Mechanisms, Contrastive Learning, Vector Embeddings, KAN, Conditional Image Generation

Computer Vision: 2D/3D Pose Estimation, Object Detection (YOLO, DETR), 3D Reconstruction, Gaussian Splatting, Kinematic Analysis, Biomechanics, Human-Object Interaction

LLM & Agentic AI: LLM Fine-Tuning, Domain-Specific Model Training, Agentic Workflows, API Integration (Claude, Gemini), Named Entity Recognition, Prompt Engineering, RAG Pipelines

Deployment & Infrastructure: TensorRT, Apple CoreML, Edge AI (Apple ARM), NVIDIA DeepStream SDK, NVIDIA TAO Toolkit, Cloud-Native Microservices, GCP Vertex AI

Languages & Frameworks: Python, JavaScript, PyTorch, TensorFlow, HuggingFace Transformers

Leadership & Research: Technical Proposal Writing, Conference Submissions, Research Team Mentorship, AI Coding Standards & Tooling, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

AI Research Engineer | **IdeasLab Formosa**, Taipei, Taiwan

May 2025 – Present

- **Pose Lifting & Motion Analysis:** Led R&D of a high-precision 2D-to-3D human pose lifting system for IdeasLab’s XView AI platform—the first markerless app to offer real-time full golf swing analysis on-device—achieving a 60% accuracy improvement by reducing mean per-joint error from 8 cm to 3 cm through temporal consistency modeling, motion-aligned lifting, and spatial refinement.
- **Research-to-Production Pipeline:** Reconstructed and adapted state-of-the-art research architectures (VGGT, TCPFormer, FreeMo, and Gaussian Splatting) into modular production components, yielding a 30% improvement in the company’s core motion understanding stack for biomechanical validity and temporal coherence.
- **Agentic LLM Development (R&D):** Architecting an agentic LLM-based AI coaching system for the XView AI app, including fine-tuning a domain-specific language model grounded in biomechanical facts and custom rule sets derived from proprietary motion analysis outputs; experimenting with API-level LLM integrations (Claude, Gemini).
- **Apple CoreML & Edge AI:** Developed and deployed AI models optimized for Apple ARM-based edge systems using CoreML, enabling 240fps on-device markerless motion tracking without cloud dependency—directly powering the XView AI iOS application used by PGA professionals.
- **AI Tooling & Team Standards:** Authored the organization’s AI coding agent rules, custom skills, and hooks (cursor-based) to standardize how the engineering team leverages AI-assisted development, driving team-wide adoption of consensus-driven AI workflows.

- **Multi-View Geometry & Benchmarking:** Integrated multi-view geometric constraints and temporal-aware lifting modules, improving 3D joint reconstruction fidelity by 30% under occlusion, fast motion, and long-tailed action distributions. Designed an internal benchmarking suite for evaluating model robustness.
- **Award-Winning Proposals & Global Representation:** Spearheaded the technical proposal and narrative that secured the “Go Healthy Taiwan” Award from TAITRA (638 proposals, 55 countries), and selected as lead engineering representative to present AI capabilities at Taiwan Expo Europe (Poland, June 2026).
- **Cross-Functional Research Leadership:** Collaborated with biomechanical experts to validate motion quality for kinematic profiling, skeleton tracking, and sport-specific movement diagnostics; led cross-functional research meetings to drive architecture selection.

AI Engineer | **President Information Corp (統一資訊)**, Taipei, Taiwan

November 2023 – February 2025

- **NVIDIA GTC 2025 Poster:** Led development of a scalable real-time planogram compliance system—presented as a poster at NVIDIA GTC 2025—combining dense object detection (YOLO) with fine-tuned vector embeddings enabling training-free product catalog updates at scale.
- **Autonomous Retail Vision AI:** Engineered a computer vision pipeline for partially occluded handheld object recognition integrating custom detection with vector-embedding search, achieving a 30% performance gain; deployed to production at Taiwan’s 8th unmanned 7-Eleven store (X8) in Keelung.
- **NVIDIA Ecosystem Integration:** Collaborated with NVIDIA to implement Metropolis microservices with generative AI across cloud-native and edge environments, leveraging TAO Toolkit, DeepStream SDK, and TensorRT for hardware-accelerated real-time inference.
- **LLM-Assisted NER:** Built an LLM-assisted Named Entity Recognition (NER) pipeline on GCP Document AI for content-aware text extraction from product imagery, reducing manual labeling overhead and streamlining cloud-native deployment.
- **Predictive Analytics:** Developed a predictive analytics model processing 5M+ data points that forecasted consumer behavior with 95% confidence intervals, directly informing targeted marketing strategies.

Software Developer | **AIBS Software Solutions**, Coimbatore, India

May 2017 – August 2021

- **Enterprise Software Engineering:** Developed custom ERP software solutions with Tally, optimizing manufacturing workflows tailored to enterprise requirements. Engineered an inventory management and tax tracking system that resolved 95% of software discrepancies and improved end-user satisfaction through full-stack development.

SELECTED RESEARCH & PROJECTS

Biomechanical-Aware Physics-Informed Pose Estimation model — *IdeasLab Formosa (2025–2026)*

- Engineered a biomechanically constrained action-specific specialized pose estimation model, proposed and authored it as a research paper, and built and deployed the model to production for golf swing movement analysis.

Content and Spatial Aware Generative Model for Inpainting — *Master’s Thesis, National Taipei University of Technology (2022–2023)*

- Proposed a GAN-based inpainting architecture integrating contextual attention in the generator and content-and-spatial attention in the discriminator, overcoming mode collapse, memorization, and underfitting in low-data regimes while extending receptive-field capacity for improved spatial consistency.

Efficient LLM Inference via Kolmogorov–Arnold Networks — *Independent Research (in progress)*

- Investigating substitution of dense MLP blocks with KAN layers to reduce LLM computational cost while preserving representational capacity, bridging mathematical function approximation theory with practical inference optimization.

Algorithm Case Studies & Transformer Architecture Breakdown — github.com/Ajishpradeep/Case_Study

- Published a rigorous mathematical deconstruction of the Transformer architecture; fine-tuned image-to-image vector embedding models using triplet loss for visual retrieval. Additional experiments with VQA, Vision Transformer depth estimation, and Vertex AI/Document AI deployments.

EDUCATION

Master of Science in Electrical Engineering and Computer Science | GPA: 3.8 / 4.0

National Taipei University of Technology, Taipei, Taiwan | 2021 – 2023

Bachelor of Science in Information Technology | GPA: 7.9 / 10

Sri Ramakrishna Mission Vidyalaya College, Coimbatore, India | 2011 – 2014

RECOGNITION & PROFESSIONAL ACTIVITIES

- NVIDIA GTC 2025 — Technical poster presenter: “Scalable Vision AI for Planogram Compliance.” Led the project end-to-end from research through conference submission.
- Go Healthy Taiwan Award (TAITRA) — Technical lead engineer behind the award-winning proposal; selected from 638 international submissions across 55 countries.
- Taiwan Expo Europe — Selected as lead engineering representative to showcase AI capabilities, Poland (June 2026).
- Event Planning Manager — International Student Association, National Taipei University of Technology (2022–2023).